Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/03/2023
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (80NSSC19K0231 ; NNX16AK21G)		
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Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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PI Email:	randgerman@gmail.com	**Fax:**	FY none
PI Organization Type:	UNIVERSITY	**Phone:**	858-922-4985
Organization Name:	San Diego State University		
PI Address 1:	College of Engineering		
PI Address 2:	282 Surfview Court		
PI Web Page:			
City:	Del Mar	**State:**	CA
Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Reyes Tirado, Fernando	**Contact Phone:**	
Contact Email:	fernando.l.reyestirado@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 3/31/2023 per NSSC information (Ed., 4/3/23) NOTE: End date changed to 6/30/2022 per PI (Ed., 8/5/21) NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 8/12/20) NOTE: Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and MSFC personnel (Ed., 2/21/19)		
Key Personnel Changes/Previous PI:	Dr. Elisa Torresani, took over student supervision starting in the summer of 2020 and is guiding both characterization and fabrication efforts as Co-Investigator. She is supervising one PhD student at this time. Previously she worked as a postdoctoral researcher on this project. She is now Assistant Professor of Mechanical Engineering at San Diego State University.		
COI Name (Institution):	Olevsky, Eugene Ph.D. (San Diego State University) Torresani, Elisa Ph.D. (San Diego State University)		
Grant/Contract No.:	80NSSC19K0231 ; NNX16AK21G		
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Task Description:

NOTE (Ed., 2/21/19): Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and Marshall Space Flight Center (MSFC) personnel.
NOTE (Ed., June 2016): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments," grant #NNX14AB31G with the same Principal Investigator.

The flight research is completed. The focus has been on the gravity-dependent densification, distortion, microstructure development, and the related macroscopic shape evolution of multi-phase powder alloys during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the liquid phase sintering induced pore-grain structure evolution by the de-convolution of the gravity impact and (ii) exploring sintering under microgravity conditions as the basis for techniques applied to in-space fabrication and repair via additive manufacturing. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit particle-based fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing based on sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. This research has developed a computer finite element code for the multiple-scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity were performed using a variety of tungsten heavy alloys (W-Ni-Cu-Mn) to validate the sintering model framework. The research outcome is relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.

At the completion of the experiments, in March 2023, all microgravity experiments were completed, consisting of 49 samples. The last 14 microgravity samples were returned to the research team a year earlier. The last 14 samples were characterized for density, mass, dimensions, and microstructure changes to document distortion, densification, and microstructure evolution. At the same time, based on the microgravity findings, new samples were fabricated for possible final ground proof experiments. A total of 14 samples, representing 2 cartridges, were sought following the C2 and C5 heating protocols. Due to expiration of the grant, the fate of any final ground proof experiments is uncertain, but sample fabrication was completed anyhow (the original ground proof samples were lost during the COVID pandemic due to a death from a support engineer in Huntsville and inadvertent disposal on cleaning his office).

This research has now been documented in several journal articles and final results are pending in two final articles pending publication in International Journal of Powder Metallurgy and Metallurgical and Materials Transactions.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, magnets, and composites. In the USA this is responsible for approximately $25 billion in annual component production. The products are applied to such diverse fields as metal cutting, oil well drilling, automotive engines, electric motors, computer heat sinks, jet engines, medical tools, skeletal implants, exercise monitors, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge in sintering since often dimensions change 15 to 20% due to shrinkage in going from the porous powder structure to the dense component. Gravity provides for a nonuniform shrinkage in height versus perpendicular directions. During high temperature sintering the component strength drops considerably and is subject to gravity induced settling effects while pores undergo buoyancy. Removing gravity allows surface energy to spheroidize both the component and pore space (agglomerated pores form a massive void). Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a verified computational model to enable prediction of final size and shape for different gravitational fields, using a multiple scale model with hopes to extend from geometry to property predictions. The robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions, as required for eventual additive manufacturing as potentially envisioned for fabrication on the moon, other planets, or in microgravity, for example, in building structure on the lunar surface or repairing space vehicles. Of course, this has benefit on Earth with the evolution of a computational means to predict high tolerance shapes by guiding sintering practice. We are adding the missing gravitational term to computer simulations, typically finite element analysis tools, for predicting final shape and size. New insight has been gained on the role of pore saturation in component shape loss, indicating that under-sintering in reduced gravity is the appropriate protocol for net-shape fabrication. A mathematical model embedded in finite element analysis is offered to predict system softening versus densification. An overarching conclusion is that ground experiments fail to provide precursor data that allows prediction of densification and distortion in microgravity. However, most likely environments with a fractional gravity will be predictable, but that was not the focus of this research.

The project has come to completion. Final findings are being published. A summary of the findings follows.
Sintering is a heat treatment applied to powders to bond the particles into a solid structure. With sufficiently high temperature, heat induces atomic motion. Through random repositioning of the solid atoms, the surface area declines as atoms move to fill high energy positions. Diffusion, especially along interfaces such as grain boundaries, is the predominant means for solid-state sintering. Faster sintering is possible if a liquid film forms on the solid grain boundaries. Atomic transport is hundreds of times faster in a liquid versus solid. Sintering with grain boundary liquid films is known as liquid phase sintering. It is a favorite for industrial production, due to the higher degree of bonding and faster rate of binding.

Liquid phase sintering is applied to the fabrication of automotive components, cutting tools, inertial weights, computer heat sinks, permanent magnets, bearings, cell phone components, diamond drilling tools, and other applications. In industrial production, the powder is shaped prior to sintering, such as by die compaction or injection molding, then sintered to produce that shape. Accordingly, sintering is frequently applied to high performance materials for a wide range of products and industries. It is especially useful for high temperature materials, such as the refractory metals.

Task Progress:

During liquid phase sintering the component is soft and viscous, allowing for densification; but also gravity, acting on the component, causes distortion. This document summarizes the findings from microgravity (µg) liquid phase sintering research conducted on the International Space Station. The study involves 49 tungsten alloy samples contained in seven cartridges. Tungsten heavy alloys are recognized for unique combinations of strength, hardness, ductility, elasticity, and high density. The common high tungsten content alloys are covered by ASTM Standard B777.

The test compositions are based on W-Ni-Cu-Mn powder mixtures. These provide high density (14 to 17 g/cm3) to enable densification and distortion experiments in small (12 mm diameter by 12 mm high) samples. The solid-liquid ratio is adjusted via changes in the tungsten content, ranging from 70 to 90 wt %. The corresponding solid:liquid ratios is about 50:50 to 80:20. In addition, at the start of the sintering cycle there is about 37 % porosity that is annihilated by sintering densification. Most of the tungsten remains solid during sintering, but undergoes significant grain growth via dissolution, diffusion, and precipitation on the larger tungsten grains. The liquid is formed from melting the admixed mechanically alloyed Ni-Cu-Mn additive. Depending on the Ni:Cu ratio, the liquid dissolves up to 28 wt % tungsten at the sintering temperature, giving the desired semisolid system. In these experiments, changes are also made to the Ni:Cu ratio from 3:2 to 2:3 to slightly adjust the tungsten solubility and dihedral angle (degree of grain boundary penetration).

Seven cartridges with seven samples each were processed in µg. One cartridge was heated to the temperature where first significant densification occurs, 1180°C, with a five minute hold. Five of the cartridges were processed for times of 10, 13, 20, 35, and 69 min at 1205 - 1210°C (a slight temperature gradient existed in the sample stack). The seventh cartridge provided a replica at 13 min at 1205°C.

Early on, this research focused on developing a W-Ni-Cu-Mn alloy to allow lower sintering temperatures to accommodate the European Space Agency downgraded Low Gradient Furnace (LGF) temperature range. The new alloys were tested using ground proof runs to gather safety, leakage, cartridge design, vibration, thermal runaway, and other data. An early failure in the quartz ampoule led to cartridge redesign. Those results provided 1g data for contrast with the microgravity data. This extensive testing documented the experimental design and verified the temperature-time profile. Subsequent runs were conducted without failure. Analysis relied on post-sintering measurements of mass, diameter (at 10 to 12 locations), height, immersion density, and microstructure parameters – solid content, solid gradient, dihedral angle, pore size distribution, and grain size distribution. A focus was on the interplay between densification (high sintered properties) and distortion (loss of shape). Modeling ideas were established and published while awaiting verification.

Sintering started at 63 % green density. Porosity remained after µg sintering. The pores were rounded and large, indicating agglomeration and coalescence, in some cases reaching up to 100 µm in diameter. The rounded pore shape arises since a subtle gas pressure stabilized the pores. Without buoyancy forces, the pores coarsen. This resulted in less densification compared to ground sintering. The pore size and grain size both follow Weibull distributions with similar median sizes (population basis). Quantitative microscopy found about one pore for every 5.4 grains. Because of the retained and coarsened porosity, a capillary force from the wetting liquid pulled the solid grains together. The resulting solid skeleton resisted distortion. Generally more liquid phase increased the sintered density, but the dihedral angle effect, from changing the Ni:Cu ratio, was not significant.

Due to a loss of ground proof samples during COVID, a new supply of 70 to 90 wt % tungsten samples were fabricated for final ground experiments. Considerable delay occurred due to personnel changes and the accompanying loss of skills. The final summary report on how microgravity influences liquid phase sintering is fairly complete, but remaining data from 14 ground samples are pending.

The µg liquid phase sintered tungsten heavy alloy compacts could not be modeled using 1g experiment data for densification or distortion behavior. Attached documents list the papers and presentations resulting from the study. [Ed. Note: See Cumulative Bibliography.] Included in those documents are details relevant to the experiments, including alloy development, sample fabrication, initial compact characteristics, flight equipment attributes, qualification testing, heating cycles, and comparison of µg and 1g results. Special attention was given to the µg thermal cycles and calculation of an integral work of sintering metric as a basis for understanding the time dependent behavior. The integral work of sintering is a means to combine heating, high temperature hold time and temperature, and cooling into a single parameter.

Residual pores abound in the sintered samples. This porosity reshaped, enlarged, and held sufficient off-gassing vapor to resist densification. Due to the capillary stress associated with the residual pores, the components remained stiff and resisted distortion. In the 1g experiments, over-sintering is observed where longer times and higher temperatures induce a loss of density. This is common to systems that produce internal gas via reactions in the pores. Thus, vapor phase pore stabilization is a significant difficulty with respect to µg sintering. In this case, the difficulty is attributed to manganese forming oxides during sintering. The oxygen is released by tungsten during the solution-reprecipitation cycle that is responsible for densification and grain growth. A related issue was a high propensity for sample wicking and adhesion to the crucible walls. A distorted shape resulted, with part of the sample curvature matching the 15 mm crucible diameter.

It is concluded that beyond thermal and green density gradients, distortion in liquid phase sintering arises due to gravity and the relatively low viscosity (3 million Pa·s) of the solid-liquid-pore system upon first melt formation. If full density is attained, then the liquid saturated space between solid grains allows for distortion. On Earth, distortion follows densification. In µg sintering, the absence of buoyancy results in porosity, pore agglomeration, pore coarsening, and a partially saturated pore structure. Accordingly, component shape takes a different trajectory. At that point, the propensity toward distortion depends on the viscosity, which in turn depends on the solid phase connectivity. Distortion in microgravity sintering requires a low liquid content (< 20 vol %) and a high dihedral angle (> 60°).

Liquid phase sintering occurs under conditions where the solid-liquid-pore system is sufficiently soft that surface energy induces viscous reshaping of the component to minimize surface area. During 1g sintering, distortion arises due to the vertical gravity body force that increases with vertical position within the component. Microgravity sintering takes a different trajectory. For example, in 1g experiments with 80W-7Ni-7Cu-6Mn at 1205°C for 60 min, the final density is 97 % in hydrogen and 92 % in vacuum. The equivalent vacuum experiment in µg delivers 89 % density. Accordingly, final component size varies due to the differences in densification. Likewise, for a 93 % W composition, there is essentially no distortion after 30 min in 1g, but 3 % distortion after 30 min in µg. Such results illustrate how sintered size and shape depends on the gravitational environment.

Distortion resistance is favored by a high solid content, high dihedral angle, and incomplete densification. Since the dihedral angle relates to the same thermodynamic parameters as wetting and solubility, a key finding is that µg liquid

	phase sintering (LPS) is more successful with low solubility systems. For WHA compositions, that implies low nickel contents. However, sintering densification is retarded by a lower tungsten solubility in the liquid matrix. Accordingly, a smaller particle size, longer hold time, or higher sintering temperature might be required to compensate. The 1g results do not provide a basis for predicting µg sintering behavior, so additive manufacturing in reduced gravity environments will most likely be unable to predict final size, shape, or properties as it will be operating without a scientific underpinning to predict densification and shape change in another gravity condition.
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